Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

April – May, 2018

CL244 Fluid Mechanics I

Date: 30.04.2018, Monday **Time: 10:00 a.m. To 1:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Fill in the blanks **[07]**

1. The flow is laminar in an open channel if the Reynolds number is less than______.
2. Droplet formation and free circular jet formation is due to ___________.
3. As roughness of open channel increases Mannings coefficient will __________.
4. Specific gravity is defined as _________.
5. Stream line and equipotential lines are _________ to each other.
6. Path of particles that passed a fixed point in flow field is known as ________ line.
7. A geometrical opening made in side or base of the vessel in such a way that water level is always above the opening is called __________.

Q - 2(a) A 45^0 reducing bend is connected in a pipeline. The diameter of inlet and outlet of the bend is 600 and 300 mm respectively. Find the force exerted by water on the bend if the intensity of pressure at inlet of the bend is 8.829N/cm^2 and the rate of flow is 600 lit/sec. **[06]**

(b) Attempt any four questions **[08]**

1 Determine the stream function given, u = 2x + y , v = x – 2. Check if the flow is continuous.

2 Differentiate between rotational and irrotational flow.

3 Obtain the expression of continuity equation for a two dimensional incompressible flow.

4 Calculate density and specific weight of one liter of petrol of specific gravity 0.8.

5 Given u = kx in a two dimensional flow determine v.

Q - 3(a) 1. A 200 mm diameter pipe conveying water branches into two pipes of diameter 150 and 100 mm respectively. The average velocity in 200 mm diameter pipe **[04]**

and 150 mm diameter pipe is 3m/s and 1.8 m/s respectively. Determine

a) Velocity in the 100mm diameter pipe and

b) Discharge carried through each three pipes in lit/sec.

2. Define the term viscosity. What is the effect of temperature on viscosity of liquids and gasses ? **[02]**

(b) Attempt any four questions **[08]**

1 Water flows over a right angled V notch at the rate 0.045 m^3/min maintaining a head of 0.048 m. Calculate coefficient of discharge of this notch.

2 The actual velocity of a liquid issuing through a 7 cm diameter orifice fitted in an open tank is 6 m/s under a head of 3 m. If the discharge measured in a collecting tank is 0.020 m3/s, calculate the coefficient of velocity, coefficient of contraction and the theoretical discharge through the orifice.

3 A venturimeter is used to measure liquid flow rate of 7500 litres per minute. The difference in pressure across the venturimeter is equivalent to 8 m of the flowing liquid. The pipe diameter is 19 cm. Calculate the throat diameter of the venturimeter. Assume the coefficient of discharge for the venturimeter as 0.96.

4 Write a brief note on flow measurement with the Rotameter

5 Derive an expression for the flow measurement in an open channel using rectangular notch.

SECTION – II

Q - 4 **Explain the following terms** **[07]**

1. Absolute pressure
2. Gauge pressure
3. Centre of buoyancy
4. Metacentre
5. Total energy line
6. Hydraulic gradient line
7. Centrifugal pump

Q – 5(a) (i) With neat sketches, explain the conditions of equilibrium for floating bodies. **[04]**

(ii) Find the volume of water displaced and center of buoyancy for a wooden block of width 2.5 m and depth 1.5 m when it is floating horizontally in water. The density of wooden block is 650 kg/m^3 and its length is 6.0m **[03]**

OR

(a) (i) State and explain Pascal's law. **[02]**

(ii A U- tube manometer is used to measure the pressure of fluid in the pipeline. The right limb of manometer contains mercury and is open to atmosphere. The contact of fluid and mercury is in left limb. Centre of pipe is 12 cm below the level of mercury in the right limb. Sketch the arrangement and find pressure of fluid in the pipe given difference of mercury in both the limbs is 20 cm. Take fluid specific gravity as 0.9. **[05]**

(b) (i) Sketch the connection of Pipes in Series and in Parallel. Write the equation of total flow through them. **[02]**

(ii) Oil at an average velocity 1 m/s flows through a pipe of 100 mm diameter. Determine whether the flow is laminar or turbulent. Also determine the friction factor and the pressure drop over 10 m length. What should be the velocity for the flow to turn turbulent? Take Density = 930 kg/m3. Dynamic viscosity μ = 0.1 Ns/m2. **[05]**

Q - 6 (a) (i) Differentiate between Pipe flow and open channel flow. **[04]**

(ii) Explain : Water Hammer effect in a pipe **[03]**

OR

(a) (i) What is meant by Most economical section of an open channel ? Obtain expression for most economical section of an open channel having Rectangular cross section **[04]**

(ii) Calculate the water discharge through an open channel shown in figure 1. Assume Chezy's constant as 60 and bed slope as 1 in 2000. **[03]**

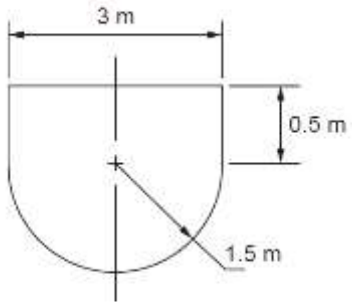

Figure 1

(b) (i) Obtain expression for Chezy's equation for velocity in open channel flow. **[05]**

(ii) Draw neat sketch of Hydraulic Ram **[02]**

****